

REVIEW

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Water saving technologies integrate agronomic environmental and digital pathways for sustainable intensification in paddy rice systems

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Abstract

Global rice production underpinning food security is increasingly constrained by freshwater scarcity and climate variability. Water-saving technologies (WSTs) such as alternate wetting and drying (AWD), the system of rice intensification (SRI), and direct-seeded rice (DSR) have been promoted to reduce irrigation demand, yet their assessment has largely focused on yield and water productivity, overlooking broader system-level effects. This review synthesizes agronomic, environmental, and digital evidence to reframe WSTs as integrated interventions for climate-resilient rice systems. Intermittent soil moisture regimes under AWD and DSR are associated with substantial methane reductions but may increase nitrous oxide emissions, resulting in context-dependent net global warming potential. Yield stability depends on synchronizing nitrogen management with aerobic soil phases, with integrated strategies improving nitrogen use efficiency. Beyond water savings, WSTs alter soil carbon dynamics, microbial processes, and irrigation-related energy use. Emerging digital tools such as in situ sensors UAV-based remote sensing, and data-driven decision models enable a shift from fixed irrigation schedules to adaptive, precision water management. These findings demonstrate that effective water-saving technologies function as integrated systems requiring synchronized management of water, nutrients, and digital tools, rather than as isolated irrigation interventions.

Keywords Alternate wetting and drying, Climate-smart agriculture, Water-saving technologies, Direct-seeded rice, Greenhouse gas emissions, Nitrogen use efficiency, System of rice intensification, UAV-based remote sensing

1 Introduction

Rice-based agroecosystems sustain more than half of the world's population yet remain among the most water-intensive forms of agricultural production. These systems are concentrated in South Asia, Southeast Asia, and East Asia, which together account for approximately 90% of global rice production and face acute groundwater depletion and climate variability challenges [1]. Irrigated lowland rice alone contributes to one-third



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of all freshwater agricultural withdrawals in the world, a dependence that has always been preserved through continuous flooding measures aimed not at resource efficiency but at stabilizing productivity [2]. This production paradigm is becoming less sustainable under accelerating stresses of groundwater depletion, climate-induced hydrologic variability, an ever-increasing energy cost, and it is also necessary to reduce agricultural greenhouse gas emissions [3]. Future rice systems must therefore be evaluated not only by yield performance but also by their capacity to maintain productivity while reducing water use, environmental externalities, and vulnerability to climatic stress. Alternate Wetting and Drying (AWD), System of Rice Intensification (SRI) and Direct-Seeded Rice (DSR) are some of the most popular alternatives to continuous flooding as water saving technologies (WSTs) [4]. In various agroecological settings, these strategies have recorded significant reduction in irrigation demand and water productivity loss usually without yield losses [5]. Nonetheless, the literature has mostly defined WSTs as irrigation efficiency interventions, and assessments were based on short-term agronomic outcomes. This tunnel vision oversimplifies system-level transformations provoked by changing soil–water regimes and understanding of their long-term agronomic ecological impacts [6].

Intermittent wetting and drying essentially reorganizes the paddy soil functionality by altering redox dynamics, the composition of microbial communities and carbon–nitrogen cycling [7]. The shifts repress methanogenesis activity, but can increase the pathways of nitrification–denitrification and create situation specific tradeoffs between methane and nitrous oxide emissions [8]. At the same time, aerobic soil phases determine nutrient availability, root development, and the dynamics of soil organic carbon, provide possibilities of resource-use efficiency and a risk of nutrient loss or carbon depletion in case management is not adjusted adequately [9]. These interactions emphasize the fact that WSTs can be characterized as system-level intervention whose effects rely on systemic water nutrient and soil management and not irrigation scheduling [10].

Recent advances in digital agriculture further expand the potential of WSTs by enabling adaptive, data-driven water management [11]. Dynamically expanded the capabilities of irrigation are now enabled by in situ soil sensors, remote sensing platform, unmanned aerial vehicle and decision support system that can relate the irrigation to the crop demand, soil moisture thresholds and also its redox conditions [12]. These feedback-driven strategies help decrease management uncertainty and provide avenues to improve the agronomic and environmental performance of AWD, SRI, and DSR. However, the integration of digital tools within water-saving rice systems has not yet been systematically synthesized in existing researches, and the associated methodologies remain poorly defined.

Previous researches have assessed water-saving technologies primarily through the lens of irrigation efficiency and water productivity, often treating them as isolated practices. This review advances beyond those assessments by constructing a systems-level framework that explicitly links water regimes to cascading effects on soil biogeochemistry, microbial communities, and plant physiology, while simultaneously evaluating the transformative role of digital feedback and policy levers. Unlike earlier syntheses, we integrate the emerging potential of AI-driven decision support, UAV-based remote sensing, and circular bioeconomy approaches as essential components of climate-smart rice

systems. Unlike earlier syntheses, we evaluate WSTs not as isolated irrigation practices but as integrated system-level interventions.

2 A conceptual framework: interlinking water, soil, plant, and data

To provide a clear roadmap for this systems-level analysis, we introduce Fig. 1 as the central organizing principle for this review. This conceptual schematic visualizes the complete causal pathway from the initial choice of water regime to the ultimate agroecosystem outcomes and governance feedbacks. Beginning at the top-left, the framework posits that the controlled Water Regime (e.g., AWD, DSR) acts as the primary driver, fundamentally altering the Soil Redox status. This redox hub governs parallel microbial, nutrient, and toxicant dynamics, which in turn drive plant and atmosphere feedbacks, including crop stress and greenhouse gas (GHG) fluxes. These biophysical processes are then quantified by digital sensing and modeling tools, whose data streams are translated into actionable insights. Finally, these insights inform targeted management and policy interventions, which close the loop by providing feedback to the initial water regime.

An integrative conceptual framework for water-saving technologies in paddy rice systems must be grounded in a systems-based perspective that captures the dynamic interactions among water management, soil physicochemical states, plant physiological processes, and data-driven decision support [13]. This framework moves beyond individual approaches to irrigation and places water redistribution, soil redox relationships, microbial ecology, nutrient conversion, and crop responses into a single causal pathway that influences both agronomic performance and environmental footprints [14]. The central driver of this framework is the management choice on WST type and timing,



Fig. 1 An integrative framework linking water management to agroecosystem outcomes and governance. This conceptual schematic synthesizes the cascading causal pathways. It begins with the controlled Water Regime, which sets the Soil Redox status—the central biogeochemical hub governing parallel microbial, nutrient, and toxicant dynamics. These soil processes directly drive plant and atmosphere feedbacks, including crop stress responses and greenhouse gas (GHG) fluxes. The framework then integrates digital sensing and modeling to quantify and attribute these processes, translating data into actionable insights. Finally, these insights inform management and policy interventions, which close the loop by providing targeted feedback to the initial water regime. The figure thus visualizes the complete cycle from driver to biophysical effect, measurement, and human adaptation, emphasizing the role of digital tools in enabling precision management and informed policy.

which alters the soil–water regime (AWD, DSR) compared to continuous flooding [15]. The detailed mechanisms (redox, methanogens) belong only in Sect. 3.1.

The agronomic and hydrological performance of water-saving technologies (WSTs) is commonly quantified using water productivity (WP) metrics, linking water inputs to biomass or yield outputs. Key formulations include:

Irrigation water productivity:

$$WP_{irr} = \frac{Y}{I} \quad (1)$$

where Y is grain yield (kg ha^{-1}) and I is total irrigation water applied ($\text{m}^3 \text{ ha}^{-1}$). This metric captures the efficiency of irrigation water use under different regimes, e.g., alternate wetting and drying (AWD) versus continuous flooding (CF).

Water productivity:

$$WP = \frac{Y}{ET} \quad (2)$$

where ET is cropping evapotranspiration (mm or $\text{m}^3 \text{ ha}^{-1}$). This reflects the crop's physiological efficiency in converting transpired water into yield and is more influenced by genotype and agronomic management.

While these metrics are essential for quantifying water productivity, it is important to acknowledge their limitations. For instance, $WP = Y/ET$ does not distinguish between beneficial crop transpiration and non-beneficial evaporative loss from the soil surface, which can vary significantly between a continuously flooded field and an aerobic one under AWD or DSR. A more nuanced interpretation of these ratios, especially when combined with measurements of evaporation and transpiration partitioning, provides deeper mechanistic insight into how different water regimes alter the water balance components and genuinely improve crop water use.

Water savings ratio:

$$WSR = \left(1 - \frac{I_{WST}}{I_{CF}} \right) \times 100\% \quad (3)$$

Where Y = grain yield (kg ha^{-1}), I = total irrigation water applied ($\text{m}^3 \text{ ha}^{-1}$), ET = crop evapotranspiration (mm or $\text{m}^3 \text{ ha}^{-1}$), I_{WST} and I_{CF} = irrigation inputs under a water-saving technology and continuous flooding, respectively.

Integrating these metrics with GHG emissions (yield-scaled GWP) allows for a multi-criteria assessment of sustainable intensification, moving beyond singular water savings to evaluate trade-offs and synergies in climate-resilient rice systems. Spatial and temporal changes in soil water status influence root architecture, aerenchyma development, and hormonal signaling. Hormones such as abscisic acid and cytokinins mediate stomatal conductance, carbon allocation, and stress tolerance. These responses connect soil hydrodynamics to plant physiological pathways under water-limited conditions [16]. The soil–plant feedback thus established determines agro-ecological results. The integrated action of moisture regime, redox control, nutrient cycles, root activity, and shoot development leads to water productivity (yield per unit water), nutrient use efficiency, and net GHG footprints [17]. As an example, AWD has been demonstrated to support or improve water productivity compared to continuous flooding and to lower methane

emissions, but N_2O effects depend on soil texture and management intensity [18]. Similarly, DSR minimizes water consumption during establishment, reduces labor inputs although it has difficulties with weeding on and seed placement which determines final yields [19].

Digital feedback systems can help close the gap between theoretical potential and practical management. Examples of continuous data streams are real-time sensor networks (soil moisture sensors and redox potential sensors), (ii) remote sensors (NDVI, thermal sensors), and (iii) predictive analytics (machine learning algorithms) that help schedule and set irrigation schedules and start points [20]. Such systems transform irrigation applications to a dynamic adaptive control of irrigation scheduling on plant water status and soil condition alarms and optimize water allocation, reduce resources waste, and stabilize yield outcomes with respect to the evolving climatic conditions [21]. Despite the absence of special development architectures of digital integration, specific contributions to preciseness water management and IoT-based feedback might be noted as a key enabler to resilient rice production systems that reconcile agronomic goals with environmental sustainability goals [22].

3 Water-saving technologies: mechanisms and systemic interactions

3.1 Alternate wetting and drying (AWD)

Alternate Wetting and Drying (AWD) extends beyond a simple water-saving irrigation practice and constitutes a fundamental transformation of paddy field hydrology that alters soil biogeochemical processes, regulates nutrient dynamics, improves soil health, and ultimately affects greenhouse gas emissions and crop performance [23]. AWD usually saves 20–35% of irrigation water compared with conventional continuous flooding (CF), and has been reported to save 25–70% of irrigation input on threshold settings and environments (not extreme drought conditions) [24, 25]. AWD reduces methane emissions by 30–70% compared to CF through inhibition of methanogenic archaea, but may increase N_2O emissions by 15–50% depending on soil and nitrogen management [26, 27]. The net climate outcome of this trade-off is explored in Sect. 4.3 and synthesized in Table 3. The resulting trade-off between repressed methane and increased N_2O determines the net GWP effect, which is extremely context dependent—determined by soil texture, organic carbon content, average annual temperature, and agricultural practices such as rate and timing of fertilizers and thereby a systemic assessment is essential to effective climate mitigation assessment [28]. A comparative summary of AWD alongside other water-saving technologies is provided in Table 2.

Nutrient dynamics under AWD are also governed by the changes in redox potentials promoting mineralization of organic nitrogen by microorganisms but also augmenting losses of nitrogen [31]. Aerobic phases enhance oxidation of ammonium to nitrate to amplify nitrification and loss of potential through nitrification and denitrification later phases, which affects the temporal synchrony between the nitrogen supply and the maximum rice demand [32]. Figure 2 presents a meta-analysis of AWD interventions, showing their effects on water productivity, methane fluxes, and N_2O emissions in paddy rice ecosystems. In terms of soil health, physical structure and relative composition of microbial population are affected by intermittent drying influence [33]. Reduced saturation decreases bulk density and alleviates compaction, thereby increasing porosity and oxygen diffusion. This shift disadvantages obligate anaerobes that dominate under

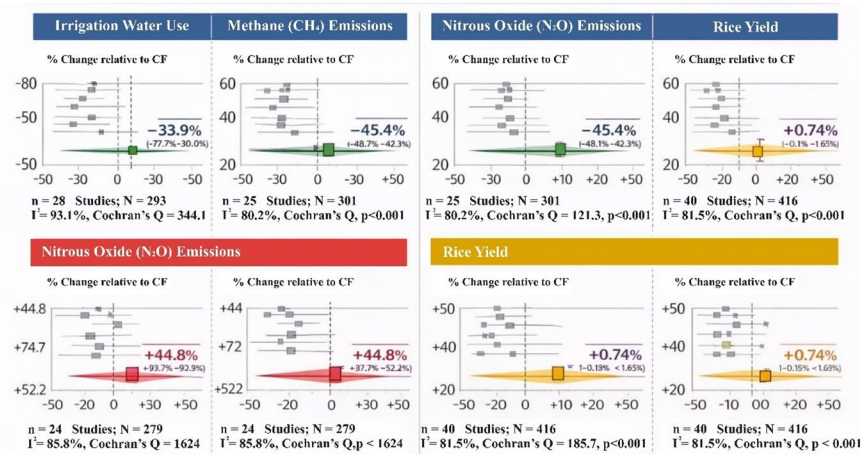


Fig. 2 The figure presents forest plots synthesize results across 24–40 independent studies (N = 279–416 observations), showing percentage changes under AWD relative to CF. AWD significantly reduced irrigation water use by 33.9% (95% CI: –77.7 to –30.0%) and methane emissions by 45.4% (95% CI: –48.7 to –42.3%), indicating strong mitigation potential. In contrast, nitrous oxide emissions increased by 44.8% (95% CI: +37.7 to +52.2%), reflecting enhanced nitrification–denitrification dynamics under intermittent soil aeration. Rice yield was statistically maintained, with a marginal increase of 0.74% (95% CI: –0.15 to +1.65%), demonstrating yield stability under AWD. High between-study heterogeneity ($I^2 = 80.2$ –93.1%; Cochran's $Q, p < 0.001$) highlights variability driven by site conditions, management intensity, and environmental context. Overall, the synthesis underscores AWD as a robust water-saving and methane mitigating practice without yield penalty, albeit with trade-offs in N_2O emissions [24, 27, 29, 30]

continuous flooding. Such transition can favor a more diverse microbiome capable of organic matter turnover and nutrient cycling, although when uncontrolled, frequent dry–wet cycles may contribute to labile carbon loss and cracking of fine-textured soils, potentially leading to preferential flow and resultant nutrient leaching. It is still unclear about the long-term consequences of soil organic carbon (SOC) stocks because less anaerobic decomposition may positively respond to stabilization but more aerobic mineralization may act as a countermeasure unless organic materials are replenished or involved in soil amendments [34].

Mechanistic AWD interactions are driven by recurrent alternations of anaerobic and aerobic soil conditions. These alternations reorganize hydrology, and microbial and plant–soil processes are strongly coupled [35]. Periodic soil drying enhances redox potential and oxygen availability repressing the activity of methanogenesis, resulting in a significant decrease in methane emissions, and instead promoting nitrification activity during periods of aerobic rewetting and denitrification activity during periods of wettings conditions; the tradeoff between the two processes is determined by the intensity of drying, soil texture, and the amount of organic matter as well as climate [36]. These redox cycles trigger change in the composition of microbial communities of obligate anaerobes to nitrifiers and aerobic heterotrophs promoting rapid mineralization of organic N but enhancing the occurrence of nitrate buildup and its subsequent loss [37]. In parallel, the moisture and oxygen cycles of the rice roots can positively affect root growth and nutrient assimilation when subjected to moderate AWD, but cause physiological stress when the drying limits are reached, and the resultant effects are highly related to stage of crop growth [38]. Nitrogen cycles are specifically sensitive, with the augmented NH_4^+ oxidation of the aerobic time frames causing an increment in leaching and denitrification losses unless substituted by enhanced fertilizer measures of deep

urea placement or controlled-release formulations, which has been demonstrated to maintain yields and enhance utilization of nitrogen [32]. These interlinked mechanisms directly operationalize the conceptual pathway shown in Fig. 1, where the AWD water regime alters soil redox status (central hub), which in turn governs microbial community shifts and nutrient dynamics, ultimately determining plant-atmosphere feedbacks including GHG fluxes. Table 1 summarizes the impacts of AWD and soil amendments on soil biogeochemistry and nutrient use efficiency (NUE).

3.2 System of rice intensification (SRI)

The System of Rice Intensification (SRI) is an integrated crop management approach in rice agronomy that combines plant, water, and soil management to amplify biological performance as opposed to adversely adding external inputs [45]. It has mechanistic bases which are based on the synergetic adjustments in planting density, water regime and soil plant interactions which together stimulate root development, minimize unnecessary competition and optimize resource capture [46]. Central to SRI's functioning is the transplanting of young seedlings at low density and wide spacing. By transplanting single seedlings at the 8–12-day stage into a square planting pattern with interplant distances typically ranging from $\sim 20 \times 20$ cm to 25×25 cm, intra-plant competition for light, water and nutrients is significantly reduced. This spacing approach allows individual plants to achieve a higher percentage of their genetic potential in terms of tillering and vegetative growth whereas enhancing deeper and extensive root development throughout the soil profile [47]. Less competition, in addition to high number of tillers per hill, enhances root length density thus increasing the size of soil volume that roots explore in search of water and other nutrients compared to normal high-density and

Table 1 Biogeochemical and nutrient use efficiency responses to AWD and soil amendments

Component	Biogeochemical and nutrient effect	Mechanism and outcomes	References
AWD redox fluctuation	Alternating oxic anoxic conditions accelerate soil organic matter (SOM) decomposition and shift carbon–nitrogen (CN) cycling. Promotes nitrification of ammonium (NH_4^+) to nitrate (NO_3^-)	Increases plant-available nitrogen (N). Mobilizes phosphorus (P) via iron manganese (FeMn) oxide destabilization	[24]
N_2O emissions under AWD	Enhanced nitrification provides more NO_3^- for denitrification	AWD increases nitrous oxide (N_2O) emissions compared to continuous flooding (CF) in many soils	[27]
Biochar organic amendment	Improves soil physical properties (porosity, water retention). Moderates rapid redox shifts and supports microbial activity	Enhances nutrient use efficiency (NUE) and water use efficiency (WUE)	[39]
Biochar nitrogen dynamics	Increases total carbon (C) and nitrogen (N), and availability of NH_4^+ and NO_3^-	Increases fertilizer-N uptake and rice yields	[40]
Urea deep placement (UDP)	Decreases ammonia volatilization and improves root access to N	Improves NUE and grain yield	[41]
Controlled-release fertilizers (CRFs)	N release is timed to match crop uptake under fluctuating moisture	Reduces N loss via leaching runoff and enhances NUE	[42]
Combination AWD + CRF	Synergistic effect on water saving and controlled N availability optimizes soil N pools and reduces microbial competition	Enhances soil N availability, rice yield, and NUE	[43]
AWD redox fluctuation	Alternating oxic anoxic conditions accelerate soil organic matter (SOM) decomposition and shift carbon–nitrogen (CN) cycling. Promotes nitrification of ammonium (NH_4^+) to nitrate (NO_3^-)	Increases plant-available nitrogen (N). Mobilizes phosphorus (P) via iron manganese (FeMn) oxide destabilization	[44]

clumped patterns of transplanting. These morphological responses have been repeatedly monitored in different environments and cultivars and have also been associated with high water-foraging ability and nutrient acquisition efficiency in SRI systems [47].

SRI's water management departs fundamentally from continuous flooding. Instead, alternate wetting and drying or maximum minimal irrigation sustains mainly aerobic conditions in soil at the vegetative stage, as opposed to the hypoxic environment, dictated by typical paddy inundation [15]. The aerobic soils facilitate the penetration of oxygen in the rhizosphere, activating the respiration of roots and maintaining the meristematic activity of roots at deeper positions within the soil profile [48]. Aerobic environments additionally increase soil microbial activity promoting positive rhizobacterial and mycorrhizal communities mobilizing nutrients and additionally benefiting root development and plant vitality [49]. Avoidance of long periods of anoxia under SRI alleviates root senescence and degradation typical of soils that are continuously flooded and therefore maintains intact functioning and extensive root systems more able to access water and labile nutrients during the crop cycle [50]. The effectiveness of AWD combined with enhanced aeration has also been observed to yield marked water saving results (significantly 30% or more) as compared to traditional flooded systems, and frequently maintains or augments grain yield [51]. It is also important to acknowledge that SRI remains a subject of ongoing scientific discourse. While numerous studies report yield advantages, the relative contribution of each management component (young seedlings, wide spacing, aerobic soils, mechanical weeding) to the final yield outcome is often difficult to disentangle, and the replication of high, consistent yields outside of well-managed research settings can be challenging. This debate underscores that the mechanistic effectiveness of SRI is not merely the sum of its parts but relies on a high degree of managerial precision and holistic adoption, which can be a limiting factor for widespread, uniform success.

Another defining agronomic mechanism in SRI is mechanical weeding, which actively aerates the soil surface. Cono-weeders or other soil loosening tools interfere with the soil surface on plant rows, enhancing gaseous exchange involving the soil and atmosphere as well as alleviating surface crusting. This facilitates root infiltration and increases the biological activity of rhizosphere, the dynamics of mineralization of nutrients and water uptake. Increased nutrient use efficiency is supported by the improved root-soil interface and has been associated with the observed increases in physiological performance characteristics based on chlorophyll content, photosynthetic rates and assimilate partitioning to reproductive organs [47]. Therefore, intensive training and design of readily available mechanical weeders and irrigation-control equipment's are essential to magnify the benefits of SRI and decreased its labor intensity under smallholder and commercial settings [52]. Table 2 presents a comparative systemic evaluation of the principal water-saving technologies.

3.3 Direct-seeded rice (DSR)

Direct-seeded rice (DSR) represents a fundamental shift in paddy rice establishment, replacing the traditional nursery-to-field transplanting sequence with direct sowing of seeds into prepared soil [58]. DSR significantly reduces water required for crop establishment by eliminating the need for nursery raising, puddling, and maintaining 4–5 cm of standing water typical of transplanted puddled rice (TPR) [59]. According

Table 2 Comparative systemic analysis of major water-saving technologies

Technology	Water savings	Yield effects	GHG emissions, methane and N ₂ O	Energy and labor	Key constraints	References
Direct-seeded rice (DSR)	Approximately 25–35% reduction in water by eliminating puddling and nursery stages Water productivity approximately 0.38–0.42 kg m ⁻³ vs ~0.31–0.38 kg m ⁻³	About 15%–70% yield increases under transplanted rice Declined yield under poor stand establishment	Methane emissions reduced up to 85–97%	Eliminates nursery, puddling and repeated flooding, Reduced labor and diesel use and Improved cost and benefit ratios	Requires high-vigor seed and integrated weed management	[53–55]
Alternate wetting and drying (AWD)	Water saving about 15–35% Irrigation reduction ~33.9% and decrease water productivity metrics	Minor decrease in yield ~1.5%	Methane decreases up to 30–70% Increase in nitric oxide emission about 42.20% Decrease global warming potential about 39%	Reduced pumping and water costs	Requires precise water-level monitoring and management	[56]
System of rice intensification (SRI)	Water savings up to 25–50%	Improves yield via optimized plant spacing and soil management	Reduced methane emission up to 35%	Management intensive with higher initial skill demand	Plant spacing and nutrient management techniques were needed	[6]
Dry/Intermittent cultivation	Water use can reduce by 53–88% vs flooded	Sustain yield up to 45%	Methane emissions significantly decreased ~70–90%	Reduced irrigation energy but needs adapted genotypes	Yield risks without adapted genotypes	[57]

to various agronomic analyses, DSR water saving systems can vary widely and mostly range between 25 to 35% less irrigation water required for TPR, although this is because waterlogging and repeated floods in seedling stages are avoided [60]. This remarkable decrease is the result of aerobic soil conditions ensured by DSR, in which soil moisture is maintained at an adequate level to control germination and initial development without prolonged flooding processes and frequent puddling interventions that characterize the conventional rice culture [61]. Cost efficiencies are also improved through the elimination of puddling and related field preparation procedures and their contribution to labor reduction and energy expenses under drought regions [62].

Mechanistically, DSR alters the soil–water regime such that the soil remains predominantly aerobic after sowing. The lack of long anaerobic periods inhibits methanogenic archaea and restricts the methane production under initial growth stages. In this review we discuss, DSR is able to mitigate total methane emissions about 60–80% or more compared to continuously flooded TPR systems and reduction up to 85–97% in

total methane emission as a function of reduced flooding depth or duration, and alleviated root physiology (increased root oxidative potential and increased oxygen release) to inhibit methanogenesis. Aerobic soil conditions also reduce reliance on diesel for puddling, lowering the energy footprint of crop establishment and downstream greenhouse gas (GHG) emissions associated with mechanized tillage [53].

Despite these resource-use benefits, the major agronomic constraint in DSR is the weed and crop establishment dynamics. Without ongoing standing water, which in TPR inhibits weed germination, weeds and rice seeds germinate equally, disadvantaging rice in the crucial early 46 weeks post-sowing [63]. Uncontrolled weeds growth at this stage can also significant reduction in yield potential and affect the water-use advantage of DSR unless controlled [64]. Combined weed management approaches, such as sequencing of the pre and post emergence herbicides such as pendimethalin and penoxsulam + cyhalofop-butyl and cultural processes have been found to significantly enhance water productivity and yield indicators through a reduction of competition towards moisture and nutrients [63]. Moreover, DSR-adapted, high-vigour genotypes, which incorporate traits like anaerobic-germinative persistence, strong initial seedling vigor, and favourable root form are essential in ensuring trustworthy stand establishment and weed competitiveness [63].

Systemically, the shift towards DSR will change the interaction network between water, soil, crop and environmental consequences. With aerobic soil conditions, water productivity is enhanced and energy use reduces and it alters greenhouse gas (GHG) flux profiles [65]. Adjustments in irrigation timing provides an opportunity to embrace precision in water management such as alternate wetting and drying (AWD) or delayed post planting irrigation to align the soil moisture with crop demand that will save on water usage without linking balance on the yield [66]. Moreover, DSR's success remains contingent on comprehensive management frameworks that address seed quality, weed ecology, and soil conditions within the broader context of sustainable intensification [63]. Table 2 presents the systematic comparative evaluation of the key water saving technologies.

4 The critical nexus: water management, soil biogeochemistry, and nutrient use efficiency

The hydrological benefits of water-saving technologies are inextricably linked to, and their long-term success contingent upon, fundamental shifts in soil biogeochemistry and nutrient dynamics. This section dissects this critical nexus, demonstrating that water savings cannot be decoupled from the management of carbon and nitrogen cycles. This nexus corresponds to the central causal chain in Fig. 1, linking the initial water regime choice through soil redox transformations to plant and atmosphere feedbacks. The periodic alterations in redox potential under AWD and the sustained aerobic conditions under DSR directly govern the cycling and availability of these essential elements, shaping both crop productivity and environmental footprints. Periodic alterations in redox potential under AWD drive carbon and nitrogen (C/N) cycling. The recurring switch between oxic and anoxic conditions causes pivotal changes in nutrient availability and mineralization [67].

Phosphorus (P) also relates to redox changes; dry and wet soil phases disrupt iron (Fe) and manganese (Mn) oxide binding capacity and dislodge P covered in mineral

complexes that consequently increase short-term availability of the plant but also predisposing P leaching and surface waters mobility [41]. Precision agronomy is demanded by this biogeochemical volatility: same processes in which nutrient availability is momentarily elevated may cause nutrient leakiness unless the time relations between nutrient release in the soil and crop uptake are achieved [68]. During AWD, a temporal dislocation between peak soil N available to and crops: temporal dislocation commonly results in SO_3 imbalanced concentrations during re-flooding events in soil, which amplifies the propensity to denitrification and yield-scaled elevated levels of N_2O compared to continuous flooding frameworks [69]. Similarly, aerobic microbial mineralization of C and N can stimulate SOM turnover reducing long-term storage of organic carbon in soil and changing soil structure and water holding capacity [70].

4.1 Microbial community reorganization and biogeochemical feedbacks

The microbial community responses to water-saving regimes are synthesized in Table 3, which details the directional shifts in key functional groups and their implications for carbon and nitrogen cycling. This functional reorganization is synthesized quantitatively in Table 3, which summarizes the directional responses of key microbial guilds under different water regimes. A critical insight from this synthesis is that the shift from CF to AWD/DSR is not merely a suppression of one group (methanogens) but a fundamental restructuring of competitive dynamics. For example, the enhanced activity of iron- and

Table 3 Responses of key soil microbial functional groups to water regimes (CF, AWD, DSR) [7, 24, 26, 27, 29, 36, 37, 46]

Microbial functional group	Response to continuous flooding (CF)	Response to AWD (intermittent flooding)	Response to DSR (predominantly aerobic)	Implications for C/N cycling & GHG emissions
Methanogenic archaea	Strict anaerobes dominate; produce methane (CH_4) from labile carbon	Periodic soil O_2 inhibits activity and growth; drying induces dormancy or shifts community composition	Lack of prolonged anaerobiosis severely limits niche and substrate availability	Suppression under AWD/DSR is the major source of GHG mitigation (30–90% CH_4 reduction)
Nitrifying bacteria (ammonia & nitrite oxidizers)	Low redox and O_2 restrict oxidation of NH_4^+ to NO_3^-	Drying cycles increase soil O_2 , promoting rapid conversion of NH_4^+ to NO_3^-	Sustained aerobic conditions favor nitrifier populations, leading to NO_3^- accumulation	NO_3^- is prone to leaching or fuels denitrification, contributing to N_2O emissions
Denitrifying bacteria	Utilize limited NO_3^- to produce N_2O and N_2 . High N_2 ratio due to low pH and abundant organic C	Rewetting events following nitrification create ideal conditions (anoxia + NO_3^-) for denitrification, creating N_2O emission hotspots	Activity depends on localized anoxia; generally lower than in AWD due to less frequent wet-dry cycles	Decoupling of nitrification (aerobic) and denitrification (anaerobic) in AWD can amplify N_2O yields
Aerobic heterotrophs	SOM decomposition is slower but favors acidogenesis	Drying-rewetting cycles cause osmotic stress but also “Birch effect” pulses of mineralization, releasing NH_4^+ and CO_2	Drive rapid SOM mineralization, releasing nutrients (N, P) but potentially depleting labile carbon pools over time	Enhanced activity under AWD/DSR can boost short-term N availability but may threaten long-term soil organic carbon (SOC) sequestration without organic amendments
Iron- and sulfate-reducing bacteria	Compete with methanogens for substrates (H_2 , acetate). Reduce Fe^{3+} to Fe^{2+}	Suppressed during aerobic phases (Fe^{2+} oxidizes). Fe redox cycling influences P sorption–desorption dynamics	Limited anaerobic periods restrict reduction processes, affecting Fe–P coupling and potentially reducing P availability	Links redox, nutrient (P) mobility, and CH_4 suppression. Fe reduction can lower CH_4 but increase P solubility

sulfate-reducing bacteria during anaerobic phases can compete with methanogens for substrates like acetate and H_2 , further contributing to methane suppression. Simultaneously, the promotion of nitrifiers during aerobic phases creates a pool of nitrate that, upon rewetting, fuels denitrifiers, creating the observed N_2O emission hotspots. Furthermore, the cycling of iron (Fe_3^+/Fe_2^+) directly influences phosphorus availability, as iron oxides can bind or release phosphate depending on the redox state. This interconnected web of microbial responses underscores that managing water regimes is, in effect, managing the entire soil microbial engine, with cascading consequences for carbon sequestration, nutrient supply, and greenhouse gas balance.

4.2 Trade-offs reduction through amendments

Organic amendments such as biochar and compost have emerged as effective modulators of soil biochemical responses in AWD regimes [34]. The beneficial properties of biochar including high surface area, porosity and alkaline buffering capacity promote soil water retention, cation exchange capacity and SOM stabilization to moderate sharp redox swings and pulses of nutrient release [71]. Experimental observations indicate that biochar supplements in intermittent irrigation can reduce N_2O emissions up to 30–50%, probably due to a combination of better root physical characteristics, mineral N adsorption, and changes in microbial communities favoring full denitrification pathways or limiting substrate to denitrifiers [39]. Biochar also enhances soil pH and macroaggregate stability and the fixation of P by soil can be restricted during dry seasons and moderate on release of sorbed P during rewetting, making biochar a more effective way of increasing P availability without causing too much loss to surface waters [39, 41].

Beyond mitigating GHG trade-offs, organic amendments like biochar and compost embody the principles of a circular bioeconomy, transforming agricultural waste streams into resources that close nutrient and energy loops [72]. Rice straw, a major residue often burned in open fields, can be pyrolyzed to produce biochar, which when returned to paddy fields enhances water retention, stabilizes soil organic carbon, and reduces N_2O emissions by up to 50% [34]. Similarly, composted farmyard manure recycles nutrients, improves soil structure, and supports microbial diversity. This circular approach not only reduces dependency on synthetic fertilizers—whose production is energy-intensive—but also lowers the energy footprint of irrigation by improving soil water-holding capacity, thereby reducing pumping needs [73]. The integration of water-saving technologies with such amendments creates a closed-loop system: saved water and energy feed back into reduced operational costs and lower emissions, while enhanced soil health sustains productivity, thereby reconciling agronomic efficiency with ecological resilience [74, 75].

4.3 NUE imperative under water-saving regimes

During drying-rewetting cycles, the conventional split method of applying urea manure is more ineffective [76]. With decreasing water and increasingly important reoxidation in the soil processes, mineral N is eventually converted and easily lost to leaching, volatilization, and denitrifying decoupling fertilizer contributions to crop demand [77]. Urea deep placement (UDP) and controlled-release fertilizers significantly enhance NUE by localizing nitrogen in the root zone and slowing its release throughout the growing season [78]. UDP, specifically, positions briquetted urea to depth (usually 7 to 14 cm) to decrease surface volatilization and temporal match-making of N availability to crop

uptake, which has been linked to improvements in grain yield and N recovery efficiency and reduction in N losses in comparison with traditional broadcast use [79]. Controlled-release fertilizers provide the same synchrony by withholding N release until soil moisture or root activity causes dissolution; thus, enhancing uptake efficiency in variable moisture environments [80].

4.4 Climate-positive vs. climate-negative AWD synthesis

The climatic performance of alternate wetting and drying (AWD), expressed as yield-scaled global warming potential (GWP), emerges as a conditional outcome governed by soil properties, baseline emission regimes, and nitrogen management intensity rather than by irrigation pattern alone (Table 3). AWD delivers climate-positive outcomes when the magnitude of CH_4 reduction exceeds any increase in N_2O [81]. This occurs predominantly in fine-textured, organic carbon-rich soils that have been flooded long-term and thus have high baseline methanogenic activity. In these systems, intermittent aeration disrupts methanogenesis and favors alternative electron acceptors [82]. In these circumstances, the percentage change in CH_4 is greater than the percentage change in nitrous oxide (N_2O), leading to a net negative change in radiative forcing. The probability of the climate-positive outcome is reinforced further when AWD is implemented in the integrated nitrogen management systems [83]. Deep placement or controlled-release fertilizers spatially and temporally decouple mineral N availability to aerobic-anoxic transitions, inhibiting short-term bursts of nitrification–denitrification suppressing N_2O emissions [84]. Similarly, stabilized amendments like biochar have been shown to stabilize redox oscillations and adsorb redox species including reactive N species which would in turn stabilize the emission peaks as well as shift the AWD to a net mitigation pathway [85]. These conditional outcomes are visualized in Fig. 4, which contrasts the economic and environmental performance of CF versus AWD-based systems.

Conversely, AWD can be neutral or even negative in systems whose potential of abatement of CH_4 is inherently small [6]. Low-organic carbon soils or rough-textured soils have low methanogenic activity under sustained flooding, and limit the potential of the CH_4 reduction subsequent to AWD adoption. Within these settings, recurrent soil drying along with large mineral N application rewards fast nitrification, nitrate buildup and consequent rewetting denitrifications silently disproportionately enhance N_2O emission in comparison to CH_4 savings [86]. Highly regulated AWD systems with high temperatures of drying or high frequency drying cycles also increase this imbalance by accelerating the aerobic mineralization and increasing the scale transient pools [4]. The combination of soil texture, organic carbon conditions, and fertilizer policy dictates whether AWD is a mitigation lever or a source of unexpected climatic trade-offs [87]. Climate-sensitive implementation thus needs predictive screening instruments, based on soil map, carbon stock, and nitrogen-controlling programs, guarantee the resulting water savings using irrigation are converted into actual recovery of net greenhouse forcing but not redistribution of CH_4 to more radiatively active N_2O . The ranges of GHG responses across technologies are summarized in Table 5, and the underlying microbial drivers in Table 3.

AWD consistently reduces CH_4 emissions by 30–70%, its effect on N_2O is highly variable, ranging from a 15% to a 50% increase relative to continuous flooding. The net climate outcome, therefore, hinges on where a specific system lies within these ranges. The

intervention becomes unequivocally climate-positive only in scenarios where the CH₄ reduction is maximized—typically in fine-textured, organic carbon-rich soils with high baseline emissions—and the N₂O spikes are effectively curtailed through integrated nitrogen management practices such as urea deep placement or the use of controlled-release fertilizers. This quantitative evidence reinforces that AWD cannot be universally categorized as "climate-smart"; its climatic performance must be diagnosed pre-implementation based on local soil properties and proposed management practices.

5 Digital revolution in precision water management

Digital technology change is fundamentally altering precision water management in paddy rice systems, whereby traditional, labor intensive tensiometers and field tubes are being replaced by integrated and data driven structures that will allow real time, spatially explicit irrigation decisions to be made [14]. Contemporary irrigation monitoring has now become the wireless Internet of Things (IoT) sensor networks, which include distributed soil moisture, temperature, water level, electrical conductivity sensors linked to cloud databases to produce continuous, field scale hydrologic profiles [88]. Detailed overview described in Fig. 3. These sensor arrays are capable of demand-based irrigation control through transmission of real time in situ measurements that only activate irrigation events once crop water thresholds are exceeded, thus significantly enhancing water productivity when compared to traditional demands [89]. These IoT enabled systems have been proven to save water and have automation capacity in agricultural situations and visual equivalents are being implemented with rice AWD and precision irrigation approaches [90] (Table 4).

Simultaneously, remote sensing and unmanned aerial vehicle (UAV) platforms can expand observation ability over sensor footprints and measures spectral patterns reflective of water state of crops and spatial variability over hectare dimensions [91]. Satellite images (e.g., ESA Sentinel 2) and high-resolution UAV multispectral imagery can be used to calculate vegetation and water indices, including the Normalized Difference Water Index (NDWI), indices of crop water stress, which can be used to relate moisture heterogeneity at the field level and the dynamic nature of plant stresses and therefore offer actionable maps to do zonal irrigation management [92, 97]. These remotely sensed products provide complementary ground-based data, making it possible to stratify management zones and variable rate irrigation prescriptions, which maximize water allocation and minimize unnecessary allocation [98]. The integration of artificial intelligence (AI) and predictive analytics represents another pillar of the digital transformation [93]. Machine learning models such as neural networks, support vector machine, and random forests can process multimodal data (weather predictions, soil sensor, historical yield, spectral indices) to predict crop water needs and the best time to irrigate. Models estimate the demand of evapotranspiration and soil moisture dynamics quantitatively, allowing irrigation plans to be proactive to changing field conditions, as opposed to calendar-based irrigation plans [99]. The integration of these tools into a decision-support framework is illustrated in Fig. 3.

Beyond monitoring, these digital tools directly inform agronomic decisions. Real-time soil moisture data enables precise AWD threshold management, preventing both excessive drying that stresses roots and premature re-flooding that increases methane emissions. Remote sensing-derived vegetation indices allow targeted nitrogen applications

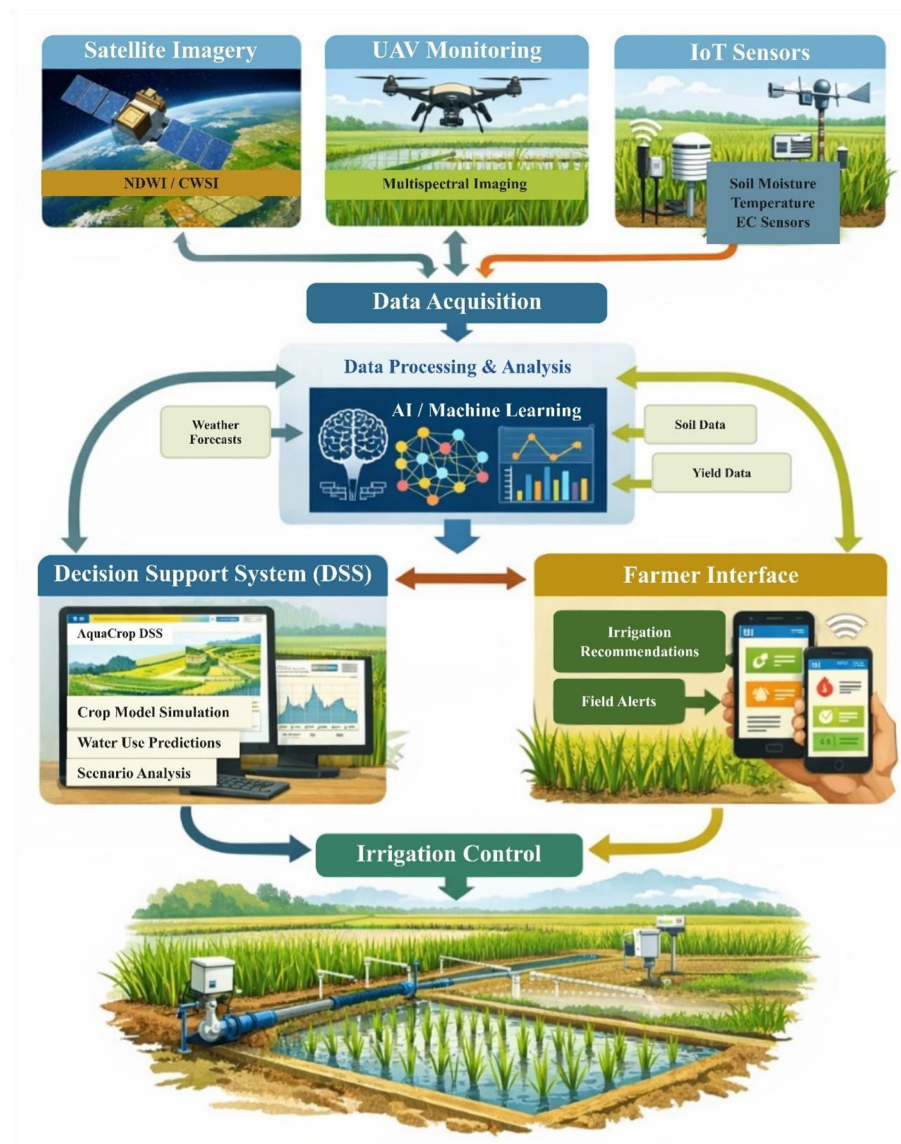


Fig. 3 The figure demonstrates conceptual framework for an intelligent irrigation DSS, structured in three layers: (i) Data acquisition (satellite/UAV imagery, IoT sensors); (ii) AI-enhanced processing and modeling (data fusion, machine learning, crop model simulation); and (iii) Decision output (automated irrigation control, predictive alerts, and farmer-facing recommendations). This integrated pipeline translates multi-scale environmental data into actionable water management

during aerobic phases when nitrification risk is highest, thereby reducing N_2O spikes. Predictive models that integrate weather forecasts with soil redox data can anticipate denitrification hotspots and recommend split fertilizer applications accordingly, synchronizing nutrient supply with crop demand while minimizing GHG trade-offs.

Underpinning these data streams are Decision Support Systems (DSS) that synthesize sensor, remote sensing, and model outputs into coherent management recommendations. Platforms such as FAO's AquaCrop and comparable crop water models can be simulated using regions-specific sensor and weather data to simulate the irrigation scenarios under different climatic conditions, enabling stakeholders to eliminate technology adoption risks and optimize irrigation strategies at farm and policy scales. Integration of wireless sensor networks, remote sensing, and DSS has improved irrigation scheduling

Table 4 Digital agriculture tools for precision water and nutrient management in rice systems

Digital tool category	Specific technologies / Data streams	Key function	Management decision supported	References
In-situ sensor networks (IoT)	Soil moisture (capacitance, TDR), redox (Eh) probes, water level loggers, ECpH sensors	Real-time, continuous monitoring of field-level hydro-biogeochemical states	Irrigation scheduling, drainage control, identifying anaerobic hotspots for targeted management	[88, 89]
Remote sensing (RS)	Satellite (Sentinel-2, Landsat) for NDVI, NDWI, LAI; UAV-drones with multispectral, thermal, hyperspectral cameras	Periodic assessment of crop status, water stress, and biomass from field to landscape scale	Zonal irrigation management, early detection of water and nutrient stress, estimating crop water requirements (ET)	[91, 92]
AI and predictive analytics	Machine learning (Random Forest, SVM, Neural Networks), time-series analysis, digital twin models	Fuse sensor, RS, and weather data to predict future crop water demand, soil moisture dynamics, and yield	Proactive irrigation planning, forecasting drought stress, optimizing planting dates	[20, 93]
Decision support systems (DSS)	Integrated platforms (AquaCrop-OS, IRRI's Crop Manager, farm management software)	Synthesize data layers into actionable recommendations (what, when, where, how much)	Prescriptive management, scenario analysis for different water-nutrient strategies, economic optimization	[94, 95]
Automation and control systems	Motorized gates, solenoid valves, cloud-connected pumps, autonomous irrigation rigs	Translate DSS outputs into physical actions with minimal human intervention	Precision application of water (variable rate irrigation), labor savings, ensuring timing accuracy	[21, 96]

accuracy, achieving water savings of 30–50% while maintaining yields, and simultaneously enabling targeted nutrient interventions that reduce N_2O emissions by synchronizing nitrogen application with soil moisture conditions [94]. Collectively, these digital advances represent a cyber physical irrigation management paradigm, in which real time sensing, spatially resolved monitoring, machine intelligence and decision automation interact to sustain sustainable intensification of rice production with precision water management as a scalable route to manage water scarcity, reduce environmental footprints, and sustain yield performance under the increasingly variable climatic regimes [95]. These digital tools constitute the 'digital sensing and modeling' layer in Fig. 1, translating biophysical processes into actionable intelligence that enables the management and policy interventions feeding back to the water regime. The key components and relationships are summarized in Fig. 3.

Despite the transformative potential of these digital technologies, their adoption in the smallholder-dominated rice systems of Asia and Africa—which produce the bulk of the world's rice—faces significant real-world barriers. The high initial cost of sensors, UAVs, and automated control systems, coupled with the need for technical literacy, reliable internet connectivity, and access to maintenance and repair services, can be prohibitive. Without addressing these constraints, the "digital revolution" risks bypassing the very farmers who could benefit most from improved water-use efficiency. Future efforts must, therefore, prioritize the development of low-cost, open-source technologies, and innovative service-delivery models (e.g., "irrigation-as-a-service" or drone cooperatives) that can democratize access to these tools. Critically, these digital tools are not just add-ons but are fundamental to operationalizing the adaptive, closed-loop management proposed in our conceptual framework (Fig. 1). They constitute the "digital sensing and modeling" layer that quantifies the biophysical processes (soil moisture, crop stress,

GHG fluxes) and translates them into actionable intelligence. This intelligence, in turn, enables the "management and policy interventions" that provide targeted feedback to the initial water regime. By enabling a shift from static, calendar-based irrigation schedules to dynamic, feedback-driven strategies, these technologies are the key enablers for managing the complex water-redox-nutrient nexus in real-time, thereby stabilizing yields and minimizing environmental trade-offs under increasingly variable climatic conditions.

6 Integration of genotype $\times$ management $\times$ environment ($G \times M \times E$)

Integration of genotype (G), management (M), and environment (E) interactions is a central component to sustainable intensification in water saving rice systems as alternate wetting and drying (AWD) and dry direct seeded rice (DSR). Classical "one-size-fits-all" breeding approaches are increasingly recognized as inadequate because rice responses to water regimes are profoundly genotype-dependent, and agronomic performance under water-saving technologies (WSTs) results from complex $G \times M \times E$ dynamics rather than isolated factors. Cultivar selection determines not only yield potential but also causes physiological and morphological adaptations that can interact with irrigation management and ambient conditions [100].

6.1 Cultivar selection and trait prioritization

AWD and other intermittent irrigation systems are subjecting high/low soil moisture and redox conditions, which demand root systems capable of sustaining water and nutrient uptake under aerobic-anaerobic transitions [101]. Phenotypic stability across water regimes is dependent on the variability in root system architecture among genotypes. For example, genotypes with higher nodal root numbers and greater root dry weight at 10–20 cm depth maintain superior water and nutrient uptake under AWD compared to conventional flooded systems [102]. AWD-dependent plasticity under genotype required effective resource acquisition. Root plasticity in dry environments (i.e. deeper rooting, increased lateral root development, and increased nutrient transporter action) is important to DSR too when seedlings have to develop highly within less saturated soils [103]. Quantitative traits loci (QTLs) in root morphology and nutrient uptake traits identified in genome wide association studies (GWAS) are strongly correlated with grain yield in dry direct seeded conditions demonstrating the genetic nature of root characteristics that provide capacity to adapt to water stress [104].

6.2 Physiological mechanisms and interactions

Under AWD, wetting and drying cycles alter soil oxygen availability, affecting root respiration and nutrient dynamics. AWD increases root activities patterns and root to shoot ratios, thereby enhancing absorption means and shrub transpiration loss, as compared to constant flooding, and leads to water productivity enhancement without effects on yield when the phenotypes are suitably accommodated to the water regime [105]. Physiological adaptations to AWD also involve hormonal changes, for example, high abscisic acid (ABA) levels, which are associated with the drought-avoidance response and the signal transduction related to the distribution of resources during intermittence [16]. Moreover, water regime and genotype are closely correlated to nitrogen uptake efficiency. Water management has an impact on the equilibrium between ammonium and nitrate state in the soil, and root systems require the plasticity to utilize variable nitrogen

status; ability to capture nitrate during aerobic periods, through a pool of nitrate transporter alleles characteristics, will enable the growth of genotypes to withstand reduced water without yield reduction [103].

6.3 High-throughput phenotyping and breeding strategies

High throughput phenotyping platforms are utilized in accelerated breeding of water saving systems to reduce and measure complex traits in the gross set of germplasm under both, controlled and field AWD/DSR treatments [106]. Digital phenotyping systems such as image-based systems, spectral sensors, and automated field systems, can be used to quickly identify genotypes with high harvest index, root plasticity and consistent productivity during periods of intermittent water stress [107]. Integrating this phenotypic information with environmental parameters and management variables would enable the selection of water saving ready varieties which is optimized to a given combination of G x M x E [108]. Practically, breeding programs combining physiological trait data, molecular markers due to root and stress tolerance loci and field-based test of water regime have provided lines with enhanced yield at semi irrigated conditions and aerobic conditions. For example, trait introgression into popular cultivars using marker assisted backcrossing has elevated root biomass and water productivity, resulting in lines that outperform recurrent parents under limited water without yield loss [109].

Field validation across South and Southeast Asia demonstrates that genotypes with increased nodal root number and root dry weight at 10–20 cm depth maintain yield stability under AWD with 15–23% water savings. Hybrid cultivars can outperform inbred lines by up to 13% under water-saving regimes. These findings underscore that breeding for root plasticity and aerobic adaptation, combined with high-throughput phenotyping, is essential for developing varieties optimized for specific water-saving environments [102, 110, 111].

7 Economic and policy dimensions

The economic feasibility and large-scale adoption of water-saving technologies (WSTs) in paddy rice systems depend not only on farm-level profitability but also on policy frameworks that correct incentive misalignments and internalize environmental externalities [112]. On-farm trials and applied economic evidence indicate that alternate wetting and drying (AWD) system has provided positive benefit–cost ratios, under farmer-operating conditions, at a range between 1.30–1.50, which represents net gains of less water, energy and input utilization and still yields sustainably comparable yield levels [113]. Despite these benefits, adoption remains inconsistent because private returns do not capture broader societal benefits, and existing water and energy pricing systems fail to incentivize conservation [114]. Traditional farm-gate valuations are likely to underestimate the value of WSTs by ignoring positive externalities, which are energy savings, mitigated methane gas emission, and enhanced food safety [115]. Periodic irrigation reduces the power needed in pumping and the cost of fuel or electricity required, and periodic soil aeration can result in a significant decrease in methane emissions when compared to continuously flooded systems. Moreover, enhanced water management has proven to lower arsenic content in rice grain, which creates the health-related benefits among customers [116]. The integration of economic and ecosystem service indicators is depicted in Fig. 4.

By adding these effects with methods of full-cost accounting such as social cost of carbon and avoided health expenditure metrics, one makes the economic argument of WSTs stronger and offers a solid argument as to why policy intercession is necessary. Multi-faceted policy tool, which addresses financial constraints, institutional constraints, and behavioral constraints, is required so that it can be scaled successfully within the smallholder-dominant landscapes [119]. Other performance-based and targeted subsidies such as more efficient irrigation equipment can mitigate the obstacles to initial investments alongside an overall more efficient utilization of water when partnered with strong performance-based extension support [120]. Reforms in the water-energy nexus are also important: an increase in the volumetric prices of water and electricity (emplacing a flat rate irrigation price system with volumetric charges) has been demonstrated to promote AWD uptake and discourage over-pumping [121]. Supplementary programs, such as ecosystem services and carbon finance, can also help private sector incentives into social goals by compensating farmers to make verifiable water savings (WS) and greenhouse gas (GHG) reductions [122]. Lastly, there is the necessity to engage citizens in extension programs through participatory extension models, including farmer field schools, community knowledge hubs, and digitally enabled advisory systems to fill the local capacity, lessen perceived risk, and feature context-specific adoption of water-saving practices [123].

7.1 Case study 1: water conservation via agronomic training

A randomized field experiment was carried out on 300 medium-scale rice farmers in groundwater-stressed districts of Haryana, India, to assess water conservation treatments under flat-fee electricity prices. The farmers were placed under a control group, an Alternate Wetting and Drying (AWD) training group, or an AWD and social comparison messaging group. AWD training incorporated face-to-face training, locally modified materials and weekly SMS pushes. The implementation of AWD also decreased pumping hours of irrigation up to 22% during the peak rates with no effect on rice yields. The highest water savings occurred in the first weeks, which is in line with AWD drying cycles. Social comparison messages were of no help than AWD training itself. The

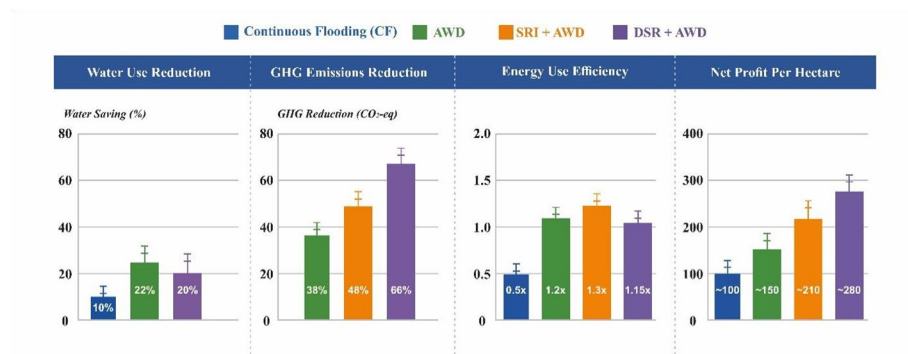


Fig. 4 The figure showed comparative assessment of continuous flooding (CF) and AWD-based rice systems integrating SRI and DSR. AWD markedly enhances system performance relative to CF, achieving ~20–25% water savings and 38–66% reductions in greenhouse gas emissions (CO₂-eq). Energy-use efficiency increased from 0.5× under CF to 1.2–1.3× under AWD-based systems. These gains translated into higher economic returns, with net profit increasing from ~100 USD ha⁻¹ (CF) to ~150–280 USD ha⁻¹ under AWD, SRI + AWD, and DSR + AWD. Overall, coupling AWD with improved crop establishment substantially improves resource-use efficiency, climate mitigation, and profitability [6, 117, 118]

findings indicate that agronomic interventions with low costs and based on knowledge can deliver high levels of groundwater conservation without reducing rice productivity [124].

7.2 Case study 2: innovative irrigation management technologies

The evidence synthesized here draws primarily from the major rice-growing regions of South and Southeast Asia, including India, Pakistan, Bangladesh, Nepal, Cambodia, and the Philippines, and China, where water scarcity and groundwater depletion increasingly constrain irrigated rice systems. These regions represent the largest concentration of rice farmers globally and are where WST adoption will have the greatest impact on food security and water sustainability. AWD consistently lowered water consumption through irrigation 23–37% and greenhouse gases 90% without yield losses and enhanced grain quality. Aerobic rice and SSC had significant water savings (27–55% and 30–40%, respectively) with moderate yield trade-offs, but SRI had water savings and yield gains under optimal management, and needed increased labor inputs. DSR minimized water consumption, labor, and energy and relied on successful weed management [14].

8 Synthesis of agronomic and environmental performance

Across diverse agroecological contexts, water-saving technologies (WSTs) consistently demonstrate that agronomic performance and environmental outcomes are jointly determined by the degree of synchronization between water regimes, nutrient management, and soil biogeochemical processes [125]. Synthesized evidence base of comparative assessments (Table 5) signifies that, under controlled conditions of the soil moisture threshold, which falls within crop sensitive limits, and nitrogen availability reflection to aerobic-anaerobic transitions, AWD, SRI, and DSR can maintain or enhance stability in the grain yields. Yield penalties reported under WSTs are not intrinsic to the technologies themselves but arise from management mismatches, particularly unsynchronized nitrogen supply during aerobic phases and inadequate weed or stand management in DSR systems [126]. With an applied integrated nutrient approach, WSTs demonstrate high water productivity and similar harvest indices to continuously flooded ones, indicating that yield formation is not dependent on high irrigation inputs [127]. Environmentally, intermittent and aerobic water regimes significantly decrease the release of methane due to a disturbance of the extended anaerobiosis, and overall climate effectiveness is dictated by simultaneous dynamics of nitrous oxide [128]. Synthesized field and meta-analytic evidence indicates that net global warming potential is largely neutral or negative under moderate AWD, provided soil organic carbon is adequate to moderate redox variations and nitrogen losses are restricted [27]. Under these conditions, the benefits of a decrease in methane take the pre-eminent role in the balance of greenhouse gas, and the benefits of climate reduction become evident without loss of productivity [129]. On the other hand, environments with coarse-textured soils or excessive nitrogen inputs exhibit greater variability in nitrous oxide responses, underscoring the need for site-specific calibration rather than uniform prescriptions. Agronomic efficiency improvements under WSTs obtained at the system level are reflected as expanded environmental co-benefits, such as the energy requirements of irrigation, ground water pumping, and soil structural conditions as compared to puddled flooding [130]. These are further improved by the digital decision-support tools where the crop reactions are

Table 5 Global ranges for agronomic and environmental indicators under water-saving technologies in paddy rice

Technology	Water savings (% reduction vs. continuous flooding)	Yield effect (% change vs. continuous flooding)	CH ₄ re- duction (%)	N ₂ O Change (%)	Key determinants of variability	Ref- er- enc- es
Alternate wet- ting and drying (AWD)	15–35	–5 to +5	30–70	+ 15 to + 50	Soil texture, organic C, N fertilizer rate, timing, drying intensity	[24, 27, 29]
Direct-seeded rice (DSR)	25–35	–10 to + 15	60–90	–20 to + 30	Weed pressure, seed vigor, soil crust- ing, post-sowing irrigation	[53, 60, 117]
System of rice intensification (SRI)	25–50	+ 5 to + 50	20–40	Not reported	Labor availability, timing of weeding and aeration, seed- ling age, soil health	[47, 51]
AWD + biochar amendment	15–50	+ 5 to + 15	40–80	–30 to –50	Biochar feedstock, soil pH, integration with N management	[34, 39]
Aerobic rice dry cultivation	40–60	–20 to + 10	70–95	Not reported	Genotype drought tolerance, rainfall reliability, subsurface irrigation	[57]

stabilized across season and management uncertainty is reduced to narrow the yield–environment trade-off space. Overall, the synthesis implies that WSTs are most effective as integrated management systems, but not such practices as individual irrigation [131]. Their environmental strength and agronomic efficiency can be actualized once the combination of the water, nutrients, soil processes, and real-time feedback systems are improved together, making the WSTs plausible solutions of sustainable intensification in rice-based agroecosystems. This synthesis of evidence across agronomic, environmental, and digital domains dispels a long-standing and persistent misconception in rice systems: that yield and environmental health are inherently locked in a zero-sum trade-off. Instead, our analysis demonstrates that the perceived conflict is, in fact, a function of management intensity. When water, nutrients, and soil biological processes are managed synergistically as an integrated system—a task now increasingly enabled by digital feedback tools—the relationship can become synergistic. The evidence compiled here shows that with precise management, it is possible to simultaneously achieve improved water productivity, reduced net global warming potential, and sustained or even enhanced grain yields. This reframing shifts the challenge from one of accepting inevitable trade-offs to one of actively designing and managing for co-benefits (Table 6).

9 Future research and policy priorities: a roadmap for scalable adoption

Future research on water-saving technologies (WSTs) must prioritize long-term, system-level evaluations. Key focus areas include soil organic carbon stabilization under intermittent redox conditions, shifts in functional biodiversity, and coupled carbon–nitrogen–phosphorus dynamics across multiple cropping cycles to inform climate-smart management. Scalable digital platforms integrating low-cost sensors with AI-enabled decision support systems are needed to translate complex soil–crop data into actionable irrigation and nutrient strategies, mitigating CH₄–N₂O trade-offs under AWD and DSR. Genetic improvement remains underexploited; breeding for root plasticity, early

Table 6 Key genotypic traits for water-saving technology-adapted rice varieties

Trait category	Specific trait	Relevance to WST (AWD, DSR, SRI)	References
Root system architecture	High nodal root number and greater root dry weight at 10–20 cm depth	Maintains water and nutrient uptake under aerobic-anaerobic transitions in AWD and DSR	[102]
	Deeper rooting, increased lateral root development	Crucial for DSR when seedlings establish in less saturated soils	[103]
	Root plasticity for resource acquisition	Enables effective adaptation to intermittent water stress	[102, 103]
Physiological adaptations	Ability to utilize nitrate (NO ₃ ⁻) during aerobic phases	Synchronizes N uptake with soil N availability under AWD, reducing N ₂ O emissions	[32, 103]
	High abscisic acid (ABA) levels for drought-avoidance response	Mediates resource distribution during soil drying cycles in AWD and SRI	[16]
	Increased root oxidative potential and oxygen release	Inhibits methanogenesis, contributing to lower CH ₄ emissions under DSR	[53]
Shoot traits	Strong initial seedling vigor	Essential for stand establishment and weed competitiveness in DSR	[63]
	Anaerobic germination persistence	Allows reliable establishment in DSR even under variable soil moisture	
	High harvest index under limited water	Maintains yield formation without high irrigation inputs	[127]
Breeding strategies	Quantitative trait loci (QTLs) for root morphology and nutrient uptake	Identified via GWAS; correlate with grain yield in dry direct-seeded conditions	[104]
	Marker-assisted backcrossing to introgress root and stress tolerance loci	Produces lines with enhanced yield under semi-irrigated and aerobic conditions without yield loss in popular cultivars	[109]

vigor, nitrate-use efficiency, and transient water-stress tolerance, combined with high-throughput phenotyping and on-farm validation, is critical for WST adaptation. Circular bioeconomy approaches that valorize rice residues into biochar, bioenergy, and biocompost can enhance water productivity, energy efficiency, and carbon balance. Life-cycle assessments and value-chain innovations are essential to support adoption and close-loop rice production systems.

To operationalize these advances, staged research-policy roadmap is required.

Short-term (1–3 years) priorities include the development and field validation of affordable digital toolkits for irrigation decision-making and simplified, region-specific models to predict net global warming potential under AWD using readily measurable soil and management indicators. Initiate participatory action research and socio-technical innovation bundling to understand and address the behavioral, cultural, and economic drivers of farmer decision-making. This includes analyzing risk perception, social network influences on technology diffusion, and co-designing adoption pathways with farming communities to ensure that new technologies are not only effective but also desirable and equitable.

Medium-term (3–7 years) efforts should concentrate on accelerated breeding for WST-adapted agroecosystems and the establishment of community-scale circular bioeconomy pilots, supported by rigorous system-level evaluations.

Long-term (7+ years) success depends on policy architectures that recognize and reward ecosystem services, including credible monitoring, reporting, and verification (MRV) systems to support payments for ecosystem services, water credits, and carbon market participation, alongside inclusive innovation platforms that enable co-learning among farmers, researchers, policymakers, and the private sector.

Achieving this ambitious vision will require a fundamental paradigm shift: from optimizing individual components of the rice system to orchestrating the entire water-soil-plant-digital-policy continuum. Only through such a holistic, systems-level approach can we transform rice cultivation from a source of environmental vulnerability into a global model of climate-smart, sustainable intensification. Collectively, water-saving technologies emerge as core components of climate-smart rice systems capable of balancing productivity, environmental integrity, and resilience under future water scarcity.

10 Concluding remarks

Water-saving technologies in paddy rice operate as integrated system-level interventions that reconfigure soil redox conditions, microbial activity, nutrient cycling, and crop physiology, collectively reshaping agronomic performance and environmental outcomes. Their effectiveness depends on coordinated water-nutrient-soil management, not on reduced irrigation alone. Greenhouse gas mitigation is strongly context-dependent, governed by soil characteristics, organic carbon status, nitrogen inputs, and the intensity of drying-rewetting cycles. When synchronized with precision nutrient management, soil amendments, and adapted genotypes, these regimes can deliver climate-positive outcomes while stabilizing yields. Digital agriculture is a key enabler for scaling adaptive, feedback-driven irrigation under climatic variability. Long-term sustainability requires the integration of genetics (breeding for root plasticity and aerobic adaptation), management, and policy, including incentive structures that value ecosystem services. For farmers, successful adoption demands integrated packages combining water-saving practices with adjusted nutrient timing, appropriate cultivar selection, and, where feasible, digital monitoring tools. For policymakers, the evidence supports reorienting subsidies from volumetric water and energy pricing toward incentives that reward verified water savings and greenhouse gas reductions, coupled with investment in extension services that build farmer capacity for precision water management.

Author contributions

Muhammad Talha Ramzan: conceptualization, visualization, supervision, project administration, writing—original draft, writing—review and editing. Laiba Razaq: writing—original draft, writing—review and editing. Zahra Noor: writing—review and editing. Saman Afzal: writing—review and editing. Asma Majeed: writing—original draft. Abdul Qadeer: writing—original draft.

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